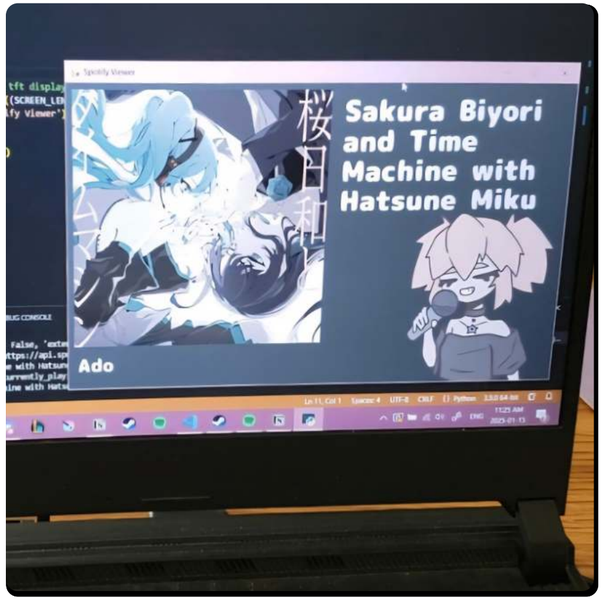
Alice Petrosyan's Portfolio

alice-petrosyan.onrender.com | a2petros@uwaterloo.ca

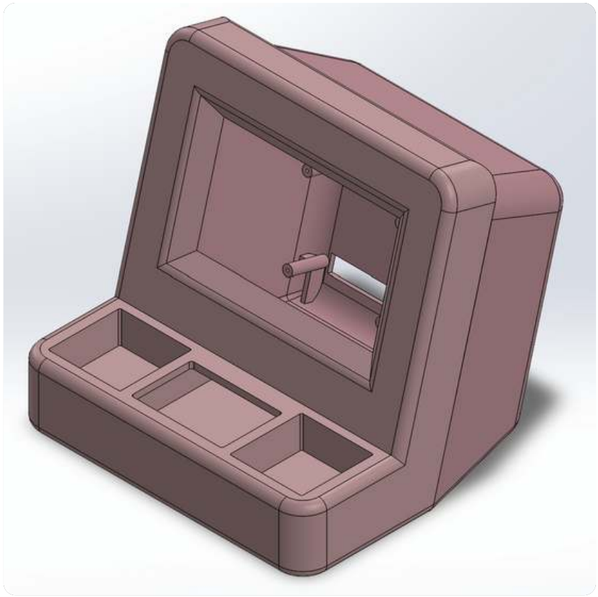
melodie

what



- A desktop tool that allows the user to view and control Spotify playback at a glance
- Features a customizable character to provide the user with a personalized look and feel, boosting the user's morale

how



- Melodie is powered by a Raspberry Pi, and coded in Python, leveraging Spotify's API
- UI and animated sprite were coded using Pygame
- Enclosure is modeled using SolidWorks

result



- Provides Spotify playback control without needing to reach for any electronic devices, minimizing distractions and improving productivity

servo depositor

what



- Designed a precision servo depositor system to address accuracy and consistency issues with existing solutions
- Created solution for food manufacturing applications requiring gentle product handling

how



- Implemented gear pump driven by Allen Bradley motor, servo drive, and PLC
- Developed dual-mode operation with customizable HMI for deposit speed, volume, timing, and user access controls

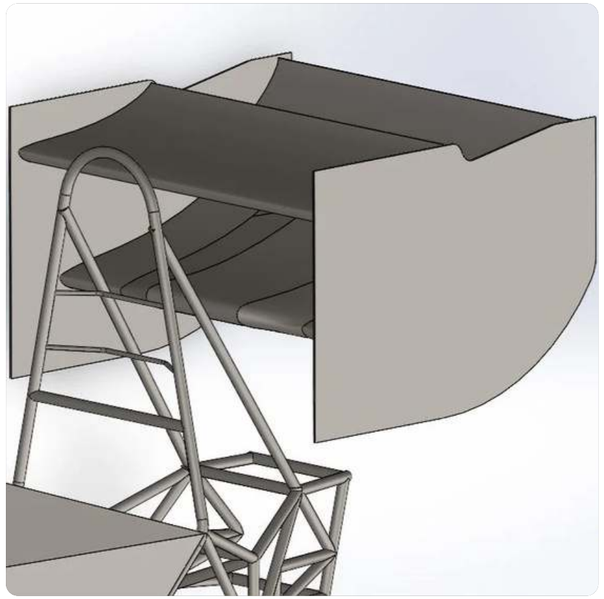
result



- Achieved consistent depositing accuracy while preserving delicate product texture
- Delivered food safety compliant system that integrates with existing pneumatic infrastructure

uwfe rear wing design

what



- The rear wing of the University of Waterloo Formula Electric's aerodynamic package for the 2026 season
- This year's goals are to increase downforce, reduce drag, and fortify mounting

how



- Lofted airfoils are designed iteratively using SolidWorks, then validated using Siemens StarCCM+ CFD software
- Swan necks and tie rods are designed using DFMA principles, while ensuring a factor of safety >2

result



- The 2026 rear wing has seen improvements in terms of efficiency (downforce vs. drag) and mounting stability
- Further analysis will be conducted after detailed design is completed

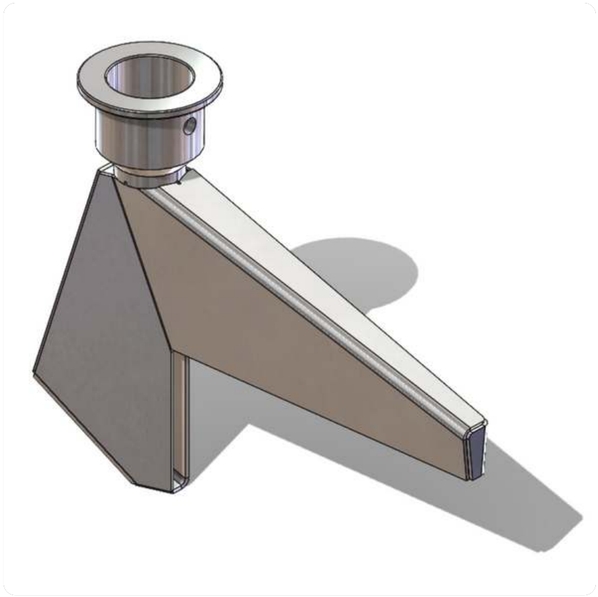
apex icing nozzles

what



- Developed specialized top and side icing (TSI) nozzles for cake decorating systems handling various cake shapes and sizes
- Addressed inconsistent frosting flow issues that resulted in uneven buttercream application

how



- Iteratively designed nozzle prototypes using SolidWorks, and validated them through 3D printing
- Manufactured final designs from sheet metal while preserving critical flow geometry from prototypes

result



- Achieved consistent frosting flow across different cake types and sizes
- Delivered smooth, even buttercream layers through optimized nozzle geometry